BMI/CEN 598 (Embedded Machine Learning) Project

SnoreGuard:
ML-Powered Pillow for snore detection and sleeping posture correction

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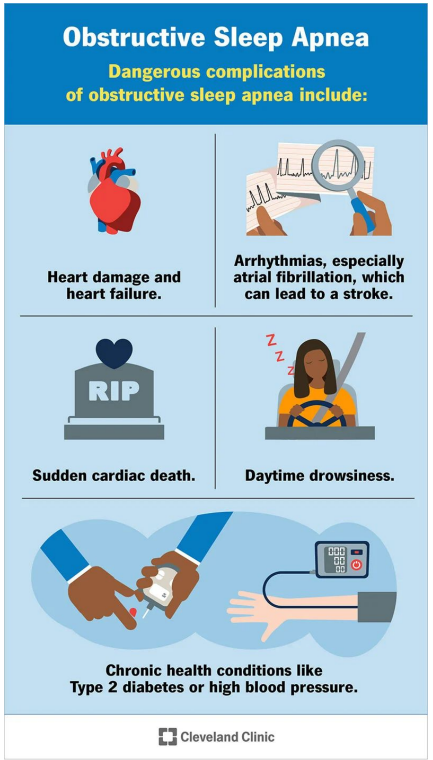
What’s in this presentation

- The problem and its significance, who cares about the problem and why?
- The state-of-the-art, what has been done before?
- System Development
- Data Processing Results
- Model Training Results
- Video Demo

The Problem and its Significance

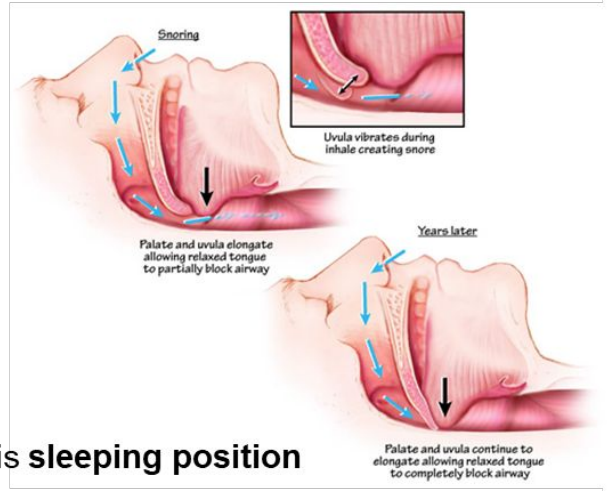
Sleep apnea is a common and serious condition that disrupts breathing during sleep, often going undiagnosed and causing various complications. Snoring is the most common form of sleep apnea and about 40% of men and 24% of women are habitual snorers. The significance of snoring as sleep apnea are as follows:

- About 75% of the people who snore suffer from obstructive sleep apnea (OSA).
- People who don't get enough sleep due to sleep apnea may have trouble staying focused and being productive, affecting their work performance.
- Untreated sleep apnea can lead to lasting physical and mental health issues over time.
- In the United States, sleep disorders result in an economic loss of $411 billion. There's a strong need for better devices to help with sleep issues, especially given busy work lifestyles.



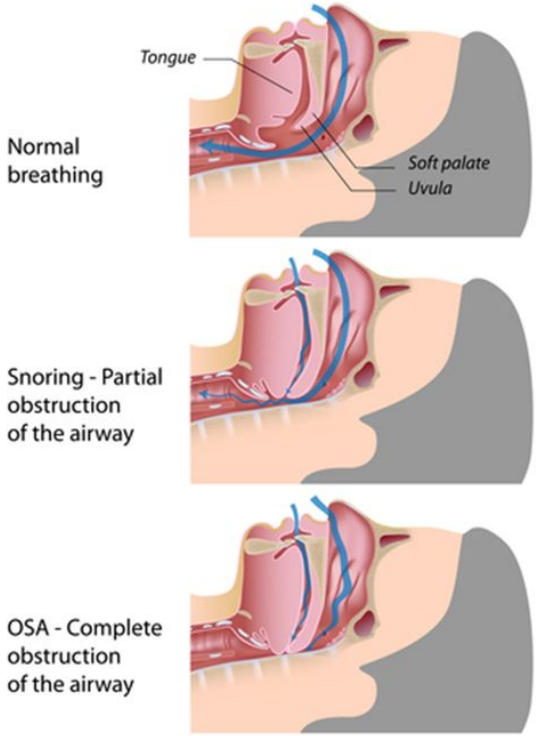
Problem: Cause

Snoring occurs when air cannot flow freely through the airway as a person sleeps.



One of the cause of snoring is **sleeping position**

When we sleep on the back, due to gravity's effect on the throat narrows the airway.



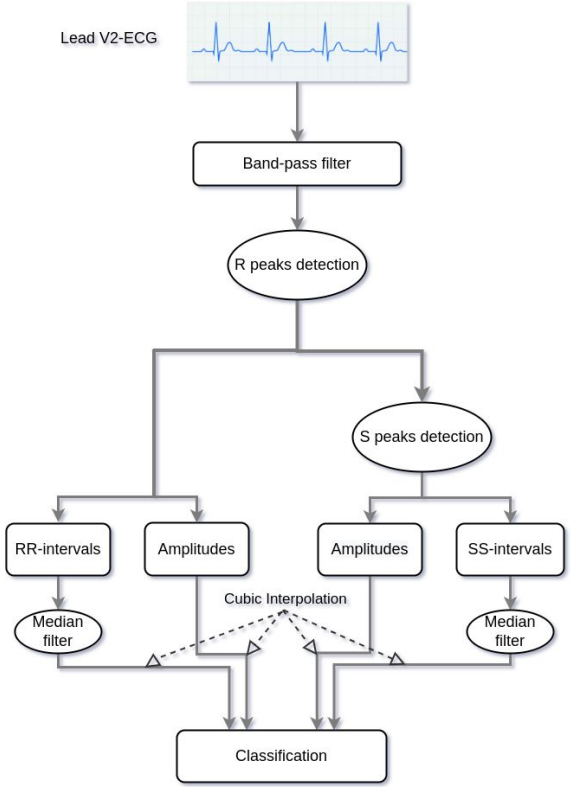
Problem: Solution

Turn the snorer from **Supine** to **Side** position

Existing simple product

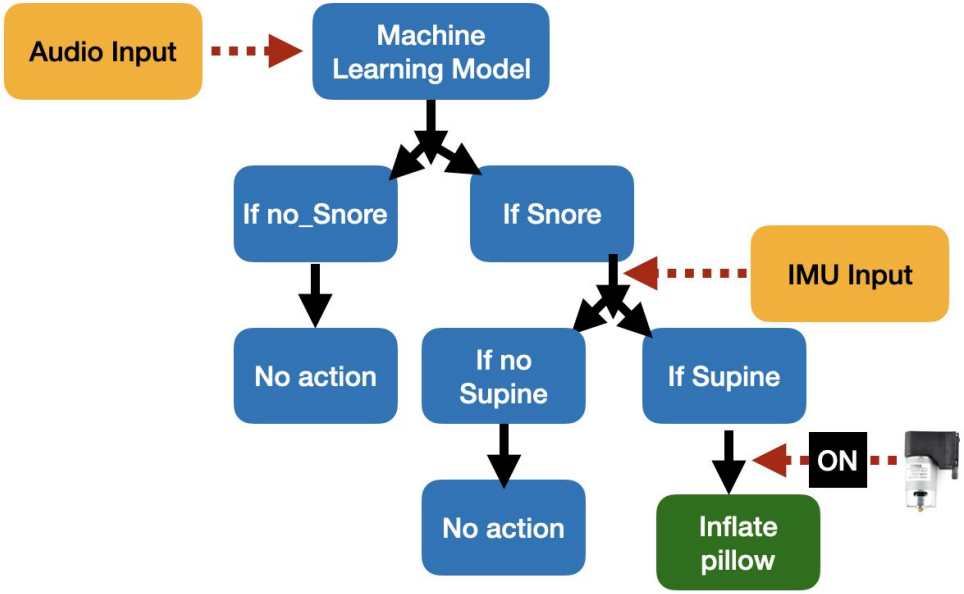
State-of-the-art SA detection methods

1. **Current Research:** Recent ECG-based SA detection has focused on feature engineering from multiple-lead ECG signals. Conformer (Convolutional + Transformer) model is the latest SOTA model.
2. **Innovation:** This study introduces a feature extraction method based on S peak detection to enhance SA segment identification using a single-lead ECG.
3. **Dataset and Lead Used:** The features are extracted from the V2 single lead and used to identify SA episodes.
4. **Machine Learning Model:** A Convolutional Neural Network (CNN) model is trained on the extracted features for SA detection.
5. **Performance:** The state-of-the-art methods have 91.13% classification accuracy, 92.58% sensitivity, and 88.75% specificity.
6. **Enhancement:** Including features associated with S peaks further improves classification accuracy by 0.85%.



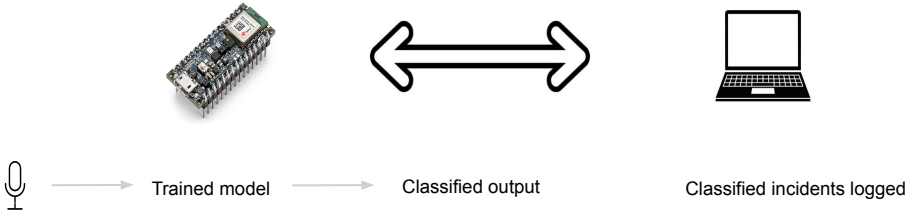
System Development

- Snore sensing Algorithm: The trained model classifies the recorded sound and labels it as snore/ not snore.
- Remote station: All the incidents of the snore are logged into a remote station , which can be used for early detection of SA.
- Receiving an alarm while sleeping can disruptive and may interrupt a person's sleep cycle.
- Our solution is different from in this aspect. Also, we will try to implement the Conformer model on the embedded system.



System Development

- **Arduino Nano 33 BLE Sense**: nRF52840 micro-controller board. In-built microphone detects snoring.
- **Relay Module 5V**: Interface between the low-power circuit of the Arduino and the high-power circuit needed for the vacuum pump. Acts as a switch.
- **Vacuum Pump 12V Mini Diaphragm Air Compressor**: Used to inflate or deflate the pillow based on the command received by the Arduino.
- **9V Battery**: To power the relay circuit.
- **Inflatable Pillow**: Inflates upon compressor activation and thus changes the position of the person.



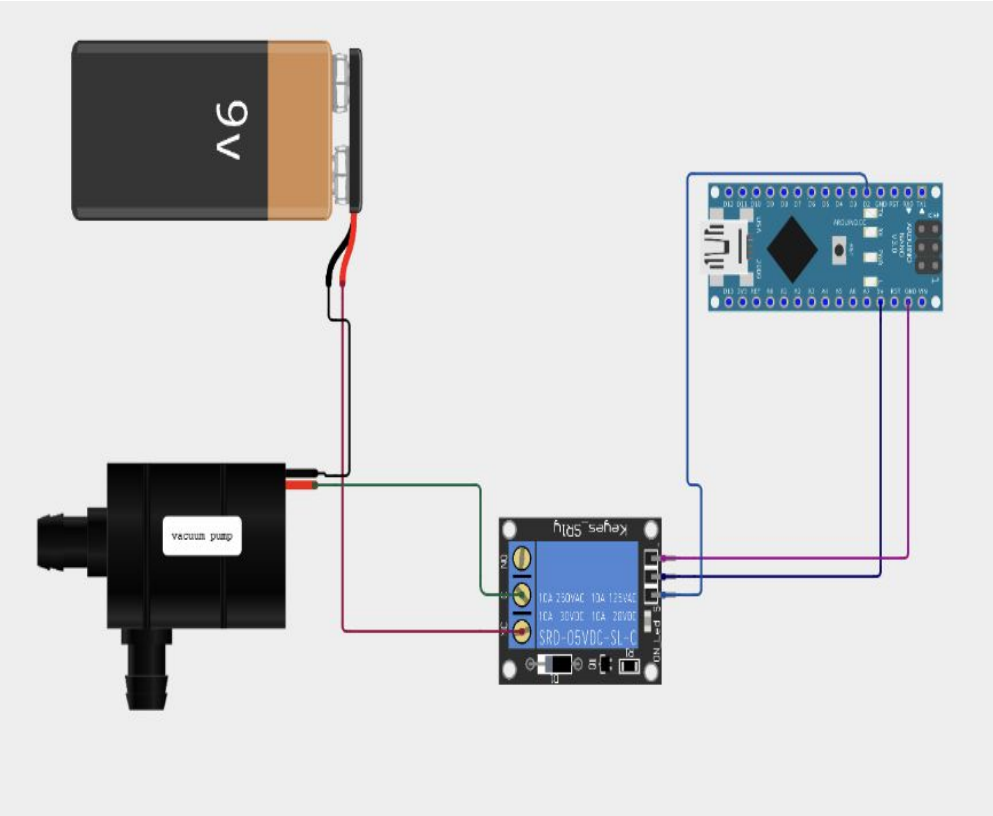
System Development

Power Distribution:
Battery positive to Vacuum Pump and Relay VCC.
Battery negative to Relay GND and Microcontroller GND.

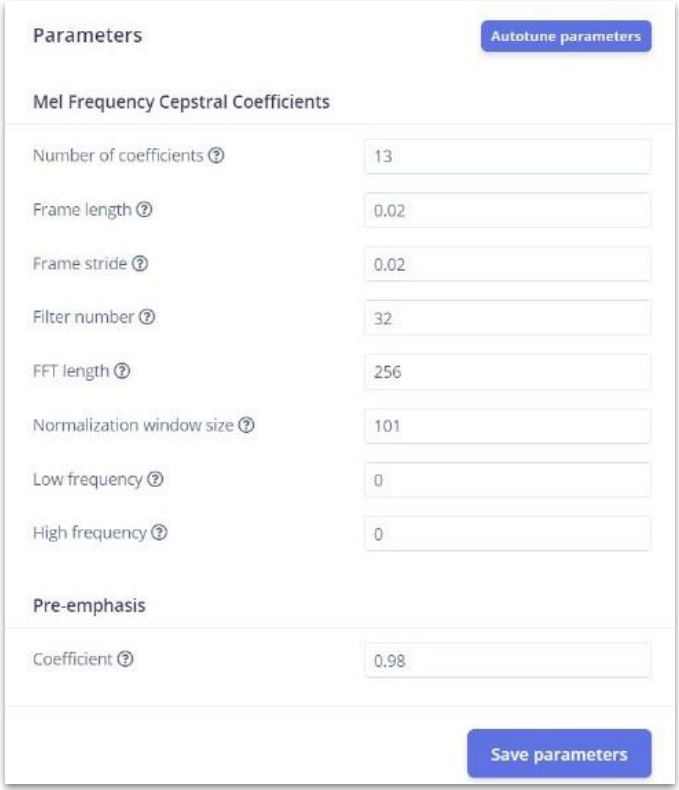
Relay Module:
COM connected to Vacuum Pump negative.
NO connected to Battery negative.
IN receives signal from Microcontroller.

Microcontroller Board:
VIN connected to Battery positive.
Digital I/O pin connected to Relay IN.

Operation:
Microcontroller sends HIGH/LOW signals to Relay IN.
Relay switches to power the Vacuum Pump on/off accordingly.



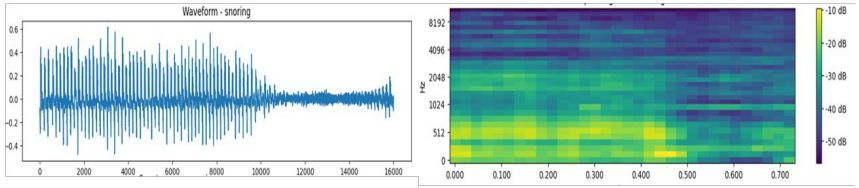
Data Processing Results



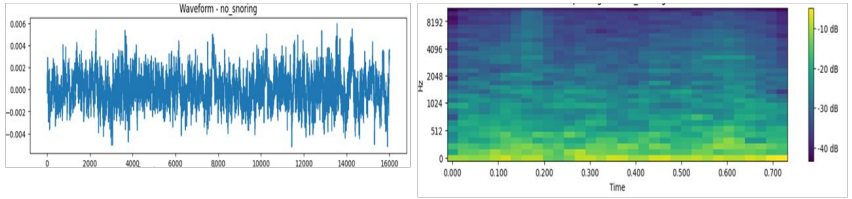
Parameters for MFCCs

Data preprocessing

Snoring



No-Snoring



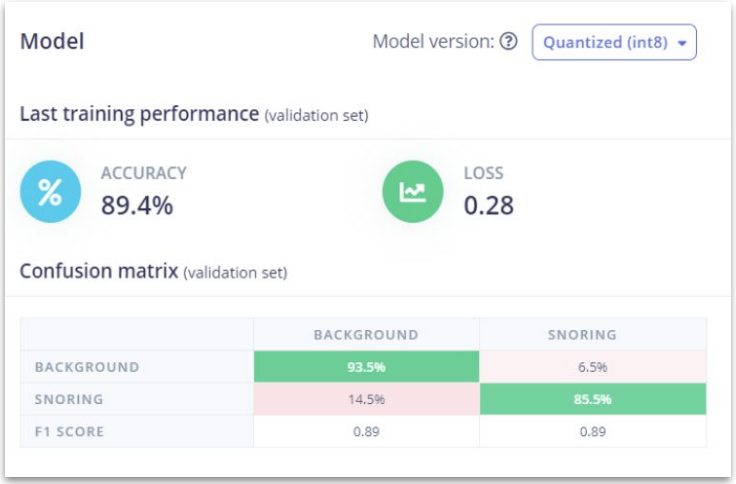
The MFCCs of a two-dimensional signal of size 32 × 32 images are created.

1D CNN Model Architecture Overview

- A Reshape layer for input adjustment.
- 1D Convolution layer with 8 filters and 320 parameters.
- Max Pooling layer for dimensionality reduction.
- A Dropout layer is implemented to prevent overfitting. Another Convolution layer with 16 filters (400 parameters).
- Max Pooling precede a further Convolution layer with 8 filters (392 parameters).
- Final Max Pooling.
- The flattened output feeds into a Dense layer indicating binary classification capabilities, totaling 1130 trainable parameters.

```
Model: "sequential_2"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 reshape_2 (Reshape)         (None, 7, 13)             0

 conv1d (Conv1D)             (None, 7, 8)              320

 max_pooling1d (MaxPooling1  (None, 4, 8)              0
 D)

 dropout (Dropout)           (None, 4, 8)              0

 conv1d_1 (Conv1D)           (None, 4, 16)             400

 max_pooling1d_1 (MaxPoolin  (None, 2, 16)             0
 g1D)

 dropout_1 (Dropout)         (None, 2, 16)             0

 conv1d_2 (Conv1D)           (None, 2, 8)              392

 max_pooling1d_2 (MaxPoolin  (None, 1, 8)              0
 g1D)

 flatten (Flatten)           (None, 8)                 0

 y_pred (Dense)              (None, 2)                 18

=================================================================
Total params: 1130 (4.41 KB)
Trainable params: 1130 (4.41 KB)
Non-trainable params: 0 (0.00 Byte)
```

Model Training Results



	BACKGROUND	SNORING
BACKGROUND	93.5%	6.5%
SNORING	14.5%	85.5%
F1 SCORE	0.89	0.89

1. Model Accuracy

2. Scatter plot for performance visualization

Video Demonstration

Contributions of Team Members

Harkirat Singh Chahal	Worked on CNN Model Architecture
Mohit Gupta	
Dheeraj Kumar	

Rohith Bacchu	Worked on the Arduino – Machine Learning Integration
Debanjalee Roy	
Harshal Wagh	

Pratik Magadwar	Worked on the circuit design
Anith Chowdary Garikapati	

The Powerpoint presentation and the manuscript was prepared by everyone.

GO DEVILS!